

KittyVerse

Where Every Paw Finds a Home

A little universe where curious whiskers meet kind hearts.

HackTheKitty 2026

Created by Team FannyPaw

GitHub Repository: <https://github.com/omichoubey793-png/KittyVerse>

Live Demo: <https://kitty-verse-lnid.vercel.app/>

Demo Video: <https://drive.google.com/file/d/1zOrxuVE70Ex7WomO56l0QsWI4bZ2frLx/view?usp=sharing>

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1. Executive Summary

KittyVerse is a modern, production-ready shelter and adoption management platform that connects rescue operations with adopters through a secure, responsive web dashboard. It brings rescue intake, medical recordkeeping, adoption review, and AI-assisted health guidance into a single connected system — turning shelter administration from a scattered, paper-heavy process into something a small team can run confidently from one screen.

Built on a modern JavaScript stack and deployed as a live platform, KittyVerse is not just a hackathon concept — it is working software. Because every paw deserves a forever home.

2. Problem Statement

Animal shelters often manage rescue intake, medical records, adoption paperwork, and donor reporting across disconnected spreadsheets and paper logs. This makes it difficult to track an animal's full journey from rescue to adoption, slows down decision-making, and leaves staff without the data they need to plan responsibly.

KittyVerse was built to close this gap — replacing scattered tools with one secure system that keeps intake records, medical history, adoption status, and shelter analytics in a single, connected place.

3. Vision & Mission

KittyVerse envisions shelters operating with the same quality of software as any modern organization — secure, data-driven, and simple to use — so that staff can spend less time on paperwork and more time on animal care. Its mission is to give administrators, staff, and adopters a single trustworthy platform for intake, medical tracking, adoption review, and AI-assisted decision support.

4. Project Overview

KittyVerse is a full-stack web application: a responsive frontend built with semantic HTML5, vanilla JavaScript, and Tailwind CSS, paired with a Node.js and Express backend and a MongoDB database. It supports rescue intake logging, vaccination and medical recordkeeping, a structured adoption review pipeline, an AI-powered health assistant, an analytics dashboard, and secure role-based authentication — deployed live, with the frontend on Vercel and the backend on Render.

5. Why KittyVerse Stands Out

KittyVerse was built as real, working software rather than a static prototype — every feature below is live in the deployed platform.

Complete Intake & Shelter Management

Staff can log rescues, assign caretakers, and track breed, age, gender, and housing assignments — keeping every animal's record centralized instead of scattered across spreadsheets.

Real Medical Recordkeeping

Vaccination logs, scheduled boosters, and clinic visits are tracked in real time, with auto-generated PDF records staff can hand to adopters or vets on the spot.

A Structured Adoption Pipeline

Applications move through a clear Pending, Approved, or Rejected flow with animal-matching criteria, replacing ad-hoc decision-making with a repeatable process.

Fanny, the AI Health Assistant

A Google Gemini-powered chatbot gives staff diet plans, recovery diagnostics, and care recommendations on demand, without needing to consult outside references.

An Interactive Analytics Bento

A dashboard surfaces donation rates, wellness index metrics, and shelter growth trends, so administrators can see how the shelter is really doing.

Secure, Role-Based Access

JWT-secured sessions with Admin, Staff, and Guest roles keep medical records and adoption decisions visible only to people who are authorized to see them.

6. Target Audience

KittyVerse is built around the people who actually run and rely on a shelter:

- Shelter administrators overseeing full operations (Admin role)
- Shelter staff managing day-to-day intake and medical care (Staff role)
- Prospective adopters and visitors browsing available cats (Guest role)
- Animal rescue organizations and NGOs seeking a digital operations system
- Donors and supporters interested in shelter impact and growth

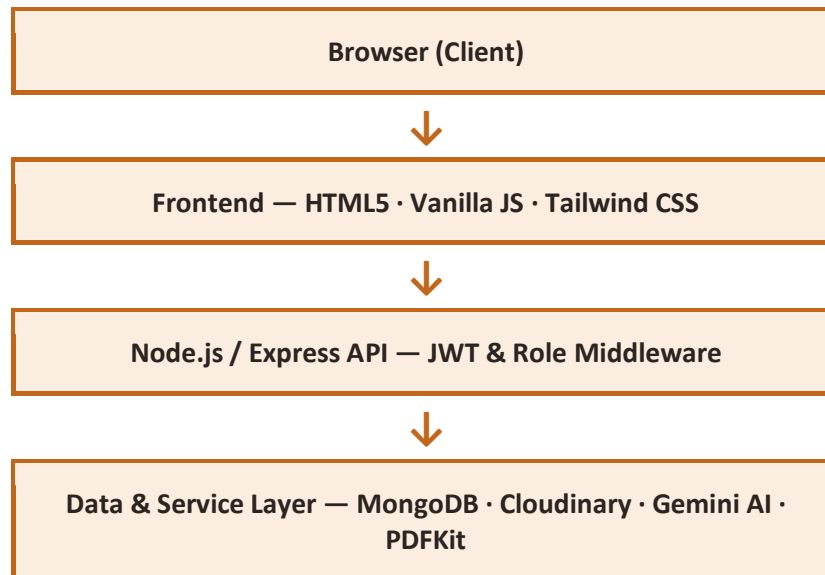
7. Core Modules & Features

- **Animal & Intake Records:** Animal profiles covering breed, age, gender, and housing assignments, created and managed by staff.
- **Adoption Management:** Applications tracked through review stages and matched against animal criteria.
- **Authentication & Roles:** JWT-secured login with Admin, Staff, and Guest permission tiers.
- **Medical & Vaccination Tracking:** Immunization schedules, clinic visit logs, and automatic PDF report generation.
- **Fanny AI Assistant:** A Gemini-powered chat assistant for health guidance and recovery diagnostics.
- **Analytics Dashboard:** Bento-style dashboard visualizing donations, wellness, and growth metrics.

8. Technology Stack

Layer	Technology
Frontend Markup	Semantic HTML5
Frontend Logic	Vanilla JavaScript (ES6 Modules)
Frontend Styling	Tailwind CSS & custom CSS tokens
Icons	Lucide Icons & Google Material Symbols
Backend Framework	Node.js & Express
Database	MongoDB (Mongoose ODM)
Media Storage	Cloudinary
AI Engine	Google Gemini API
Document Generation	PDFKit
Authentication	JSON Web Tokens (JWT)

9. Technical Architecture



Repository Structure

- **Frontend:** assets, css, and js folders (including config.js for environment routing, plus sidebar, login, and chatbot modules) and index.html as the public landing page.
- **Backend:** a config folder for MongoDB and Cloudinary setup, controllers for API endpoints, middleware for JWT and role verification, models for User, Animal, and Adoption schemas, routes for Express mappings, and server.js as the API entry point.

This modular structure separates presentation, business logic, and data storage, making the platform easier to maintain, secure, and scale.

10. Security & Access Control

Every session in KittyVerse is secured with JSON Web Tokens, and dedicated middleware enforces three access tiers — Admin, Staff, and Guest. This keeps medical records, adoption decisions, and internal analytics visible only to authorized users, while adopters and visitors still get a simple, guest-friendly browsing experience.

11. Deployment & Hosting

KittyVerse runs as two independently deployed services. The frontend is hosted on Vercel using a root-level routing configuration, while the Express backend is deployed as a web service on Render. This

separation keeps the static site fast and globally distributed while letting the API scale on its own. The live platform is available at kitty-verse.vercel.app.

12. Social Impact

By bringing intake, medical care, and adoption review into one accessible system, KittyVerse helps shelters and NGOs move animals through the adoption pipeline faster and with better-documented care. The analytics dashboard also gives shelters visibility into donation trends and wellness metrics, helping them plan more effectively and show their impact to donors and the community.

13. Learnings

Building KittyVerse deepened the team's understanding of full-stack architecture, secure authentication and role-based access design, integrating third-party AI and cloud services such as Google Gemini and Cloudinary, and the practical realities of deploying a two-service application across Vercel and Render.

14. Conclusion

KittyVerse is more than a hackathon idea — it is a working, deployed platform with real infrastructure behind it: a secure Node.js and MongoDB backend, an AI-assisted health assistant, and a live production deployment that shelters and adopters can use today.

It shows what happens when compassion is paired with solid engineering.

Because every paw deserves a forever home.